CHAPTER ONE

1.1 Background to the Study

The use of antibiotics in poultry production has become an integral practice for disease control, prevention, and growth enhancement. However, the widespread and often indiscriminate application of these drugs has contributed to the emergence of antibiotic-resistant bacteria within poultry production systems. Poultry manure, a major by-product of this industry, is frequently applied to agricultural land as organic fertilizer, thereby serving as a potential reservoir for antibiotic residues and resistant bacterial populations. Studies have shown that poultry waste harbors diverse microorganisms, including pathogenic species such as *Escherichia coli*, *Salmonella*, etc. Many of which exhibit varying degrees of resistance to commonly used antibiotics (Ngogang *et al.*, 2020).

The environmental implications of this are significant, as the application of untreated manure can facilitate the dissemination of resistant bacteria and resistance genes into soil and water systems, posing risks to both ecosystem integrity and public health.. As a response to these concerns, anaerobic digestion using batch biogas reactors has gained attention as a sustainable waste management strategy. This process not only produces renewable energy but also generates digestate, which is often utilized as a soil amendment. Although anaerobic digestion has been reported to reduce microbial load, evidence suggests that it may not completely eliminate pathogenic or antibiotic-resistant bacteria, with a little percentage of rganisms surviving the digestion process (Ndubuisi-Nnaji *et al*., 2018).

Furthermore, the persistence of resistant bacteria in digestate raises critical questions regarding its safety for agricultural use and its potential role in the continued spread of antimicrobial resistance. Antibiotic susceptibility testing remains a key approach for determining sensitivity patterns and play a role in mitigating microbial risks (Yakobi and Nwodo, 2025). Therefore, assessing and comparing the antibiotic sensitivity patterns of bacterial isolates from untreated poultry manure and digestate from batch biogas reactors is essential. Such evaluation will provide insight into the efficiency of anaerobic digestion in reducing antibiotic resistance and contribute to improved waste management practices and also help us gain insight on the things we consume daily.

1.2 Statement of the Problem

The rapid emergence and spread of antibiotic-resistant bacteria have become a major environmental and public health concern worldwide. In poultry production systems, antibiotics are frequently used for therapeutic, prophylactic, and growth-promoting purposes. This widespread usage has led to the development of resistant bacterial strains that are subsequently excreted in poultry manure. When such untreated manure is disposed of or applied to agricultural land, it serves as a pathway for the introduction of antibiotic-resistant bacteria into the environment. Evidence shows that poultry waste contains diverse bacterial populations with varying resistance to commonly used antibiotics, thereby increasing the risk of environmental contamination and human exposure (Ngogang *et al.*, 2020).

The environmental impact of this practice is further amplified by the ability of resistant bacteria and their associated resistance genes to persist and spread through soil, water, and crops. This creates a continuous cycle of contamination that may ultimately affect food safety and public health. Studies have highlighted that animal waste is a significant contributor to the dissemination of antimicrobial resistance in the environment due to excessive use of antibiotics, emphasizing the need for effective management strategies (Ngogang *et al.*, 2020).

Anaerobic digestion using batch biogas reactors has been proposed as a sustainable solution for managing poultry waste, as it reduces organic load while producing renewable energy. However, despite its advantages, there is growing concern regarding its effectiveness in completely eliminating antibiotic-resistant bacteria. Research indicates that although anaerobic digestion can significantly reduce microbial populations, some pathogenic and indicator organisms may survive the process and remain in the resulting digestate (Ndubuisi-Nnaji *et al.*, 2018) . This raises critical questions about the microbiological safety of digestate when applied to agricultural soils.

Moreover, there is limited information on the comparative antibiotic sensitivity patterns of bacterial isolates from untreated poultry manure and digestate, particularly under batch biogas reactor conditions. Without such data, it is difficult to ascertain whether anaerobic digestion sufficiently mitigates the risk of antibiotic resistance dissemination. Therefore, this study seeks to address this gap by evaluating and comparing the antibiotic sensitivity patterns of bacterial isolates from these two waste forms, thereby providing insights into their potential environmental and public health implications.

1.3 Research Questions

This study is designed to address key questions regarding the occurrence and behavior of antibiotic-resistant bacteria in untreated poultry manure and digestate from a batch biogas reactor. The following research questions will guide the study:

1. What types of bacterial isolates are present in untreated poultry manure and digestate obtained from a batch biogas reactor?
2. What are the antibiotic sensitivity patterns of bacterial isolates obtained from untreated poultry manure?
3. What are the antibiotic sensitivity patterns of bacterial isolates obtained from digestate?
4. Does the process of anaerobic digestion significantly reduce the prevalence of antibiotic-resistant bacteria?
5. Is there a significant difference in the antibiotic sensitivity patterns of bacterial isolates from untreated poultry manure and those from digestate?

1.4 Objective of the Study

The main objective of this study is to evaluate the antibiotic sensitivity patterns of bacterial isolates obtained from untreated poultry manure and digestate from a batch biogas reactor, with a view to determining the effectiveness of anaerobic digestion in reducing antibiotic-resistant bacteria.

To achieve this aim, the study will pursue the following specific objectives:

1. To isolate and identify bacterial species present in untreated poultry manure and digestate
2. To determine the antibiotic sensitivity patterns of bacterial isolates from untreated poultry manure using standard susceptibility testing methods
3. To determine the antibiotic sensitivity patterns of bacterial isolates from digestate obtained from a batch biogas reactor
4. To compare the antibiotic resistance profiles of bacterial isolates from untreated poultry manure and digestate
5. To assess the extent to which anaerobic digestion influences the reduction or persistence of antibiotic-resistant bacteria
6. To evaluate the potential public health implications associated with the use of untreated poultry manure and digestate as soil amendments

1.5 Research Hypothesis

- **Null Hypothesis (H_0):** There is no significant difference in the antibiotic sensitivity patterns of bacterial isolates obtained from untreated poultry manure and those obtained from digestate produced by a batch biogas reactor.
- **Alternative Hypothesis (H_1):** There is a significant difference in the antibiotic sensitivity patterns of bacterial isolates obtained from untreated poultry manure and those obtained from digestate produced by a batch biogas reactor.

1.6 Significance of the Study

The growing concern over antibiotic resistance has made it necessary to critically evaluate common waste management practices in livestock production systems. This study is significant as it provides insight into the role of poultry manure and digestate as potential reservoirs of antibiotic-resistant bacteria. By assessing and comparing the antibiotic sensitivity patterns of bacterial isolates from untreated poultry manure and digestate, the study contributes to a better understanding of how anaerobic digestion influences microbial safety.

The findings of this research will be valuable to environmental scientists and microbiologists by providing empirical data on the effectiveness of batch biogas reactors in reducing antibiotic-resistant bacteria. It will also assist in bridging existing knowledge gaps regarding the microbiological quality of digestate, especially in relation to its use as an organic fertilizer. This is important because previous studies have indicated that although anaerobic digestion reduces microbial load, some resistant and potentially pathogenic organisms may persist in the final digestate (Ndubuisi-Nnaji *et al*., 2018)..

Furthermore, the study will be beneficial to farmers and agricultural practitioners who rely on poultry manure and digestate for soil fertility improvement. Understanding the potential risks associated with these materials will encourage safer handling practices and informed decision-making in their application. The research also has public health relevance, as it highlights the potential pathways through which antibiotic-resistant bacteria can enter the food chain and the wider environment through agricultural practices (Ngogang *et al.*, 2020).

In addition, regulatory bodies and policymakers may find the results useful in developing guidelines for the safe use of animal waste and digestate. Overall, this study contributes to sustainable waste management practices and supports efforts aimed at mitigating the spread of antibiotic resistance.

1.7 Scope of the Study (Delimitation)

This study is focused on evaluating the antibiotic sensitivity patterns of bacterial isolates obtained from untreated poultry manure and digestate produced from a batch biogas reactor. The research is limited to samples collected from selected sources, and the analysis will be confined to culturable bacteria present in these samples.

The study will involve the isolation and identification of bacterial species using standard microbiological techniques. Antibiotic sensitivity testing will be carried out using commonly prescribed antibiotics and established methods to determine the susceptibility or resistance patterns of the isolates. The comparison will be based on the differences observed between bacterial populations in untreated poultry manure and those present in digestate after anaerobic digestion.

Furthermore, this research is restricted to the assessment of phenotypic antibiotic resistance patterns and does not extend to molecular characterization of resistance genes. Environmental factors such as seasonal variation, geographical differences, and long-term field application effects are beyond the scope of this study.

The study also focuses specifically on a batch biogas reactor system and does not cover other types of anaerobic digestion systems such as continuous or plug-flow reactors. Overall, the scope is designed to provide a controlled and comparative evaluation of antibiotic resistance in the selected waste materials without extending into broader environmental or molecular analyses.

1.8 Limitations of the Study

One major limitation is the inconsistent electricity supply, which affects the ability to maintain optimal and stable temperature conditions required for effective anaerobic digestion in the batch biogas reactor. Temperature is a critical factor influencing microbial activity and pathogen reduction; therefore, fluctuations may impact the efficiency of the digestion process and the extent to which antibiotic-resistant bacteria are reduced in the digestate.

In addition, the study is constrained by limited access to advanced molecular techniques that could provide more detailed information on antibiotic resistance genes and their mechanisms of transfer. As a result, the study focuses only on phenotypic antibiotic sensitivity patterns.

Furthermore, variations in sample composition due to differences in poultry feed, management practices, and environmental conditions may introduce some level of variability in the results. These limitations are acknowledged as factors that may influence the generalization of the findings, but they do not undermine the overall validity and relevance of the study.

1.9 Operational Definition of Terms

1. Antibiotic Sensitivity: The degree to which bacterial isolates are susceptible to the inhibitory or bactericidal effects of specific antibiotics, as determined through standard laboratory testing methods.
2. Antibiotic Resistance: The ability of bacteria to survive and grow in the presence of antibiotics that would normally inhibit or kill them.
3. Bacterial Isolates: Pure strains of bacteria obtained from poultry manure and digestate samples through laboratory culturing techniques.
4. Poultry Manure: Waste material (droppings) produced by poultry birds, which may contain microorganisms, antibiotic residues, and organic matter.
5. Digestate: The semi-solid or liquid by-product obtained after the anaerobic digestion of organic materials in a biogas reactor.
6. Anaerobic Digestion: A biological process in which microorganisms break down organic matter in the absence of oxygen, resulting in the production of biogas and digestate.
7. Batch Biogas Reactor: A type of anaerobic digester where organic material is loaded into the system and allowed to decompose over a period of time without continuous input or output.
8. Antibiotic Susceptibility Testing: Laboratory procedures used to determine the response of bacterial isolates to specific antibiotics, commonly performed using methods such as disk diffusion.
9. Pathogenic Bacteria: Disease-causing bacteria that may be present in poultry manure or digestate and pose risks to human and environmental health.